TD SheeT 4

exercise 01

1- $\Omega = \{FF, MM, FM, MR\}$

$A = \{FF\}$

$B = \{MM\}$

$C = \{FM, MF, MM\}$

$A \cap B = \emptyset$

$A \cup B = \{FF, MM\}$

$A \cap C = \emptyset$

$A \cup C = \Omega$

$B \cap C = \{MM\}$

$B \cup C = \{MM, MF, FM\}$

$C \cap B^c = \{FM, MF\}$

$B^c = \{FM, MF, FF\}$

exercise 02 :

1- $(A \cap B) \cup (A \cap B^c) \overset{?}{=} A$

$(A \cap B) \cup (A \cap B^c) = A \cap \underbrace{(B \cup B^c)}$

$= A \cap \Omega$

$= A$

2- If $B \subset A$ Then $A = B \cup (A \cap B^c)$

we have: $(B \cup A) \cap (B \cup B^c) = (B \cup A) \cap \Omega$

Since $B \subset A$ Then : $B \cup A = A$

Then $A \cap \Omega = A$

3- showing That $(A \cap B)$ and $(A \cap B^c)$ are mutually exclusive:

means That : $(A \cap B) \cap (A \cap B^c) = \emptyset$

we have :

$(A \cap B) \cap (A \cap B^c) = (A \cap A) \cap (B \cap B^c)$

$(B \cap B^c) = \emptyset$ $\Rightarrow A \cap \emptyset = \emptyset$

$(A \cap A) = A$

exercise 03 :

1- $p(\Omega) = 1$.

and $p(\Omega) = p(E_1) + p(E_2) + p(E_3) + p(E_4) + p(E_5)$

Since $1 = E_1 \cup E_2 \cup E_3 \cup E_4 \cup E_5$

Then $1 = 0{,}15 + 0{,}15 + 0{,}4 + 3\,p(E_5)$

$p(E_5) = \frac{1 - 0{,}3 - 0{,}4}{3} = 0{,}1$

$p(E_4) = 2 \times 0{,}1 = 0.2$

2- $p(E_3) = p(E_4) = p(E_5)$

$p(\Omega) = p(E_1) + p(E_2) + p(E_3) + p(E_4) + p(E_5)$

$= 4p(E_1) + p(E_3)$

$p(E_3) = \frac{p(\Omega) - 4p(E_1)}{3} = \frac{1 - 0{,}4}{3} = 0{,}2$

Since $p(E_3) = \frac{0{,}3}{3} = 0{,}1$

exercise 04 :

First we need To calculate The $|\Omega|$

$|\Omega| = \binom{50}{3} = 19600$

1- Organize all prizes :

$p(A) = \frac{\binom{4}{3}\binom{46}{0}}{\binom{50}{3}} = \frac{4}{19600}$

2- exactly Two prises :

$p(B) = \frac{\binom{4}{2}\binom{46}{1}}{\binom{50}{3}} = \frac{276}{19600}$

3-

$p(C) = \frac{\binom{4}{1}\binom{46}{2}}{\binom{50}{3}} = \frac{4140}{19600}$

4-

$p(D) = \frac{\binom{46}{3}\binom{4}{0}}{\binom{50}{3}} = \frac{15180}{19600}$

Exercise 05:

1. Ω : pick 3 randomly.

$|\Omega| = \binom{18}{3} = 816$

A : All the three balls are not of the same color: $\binom{6}{1}\binom{4}{1}\binom{8}{1} = 6 \times 4 \times 8 = 192 = |A|$

$p(A) = \frac{|A|}{|\Omega|} = \frac{192}{816} = 0,24$

2. G : both are green

$p(G_A) = \frac{3}{10}$, $p(G_B) = \frac{10}{15}$

$p(G) = \frac{3}{10} \times \frac{10}{15} = 0,2$.

Exercise 06:

1. The number of different samples of 10 can be selected $\binom{90}{10} = 5,72 . 10^{12}$

2. The number of ways fo selected 4 males $\binom{20}{4}$

The number of ways for selected 6 females $\binom{70}{6}$

So the probability is: $\frac{\binom{70}{6} \times \binom{20}{4}}{\binom{90}{10}} = 0,11$

Exercise 07:

1. We have 5 firms and 3 contracts

$|\Omega| = 5 \times 4 \times 3 = 60$

2. Finding the probability:

Let A be the event

$p(A) = \frac{A_2^4 \times 3}{60}$

$= \frac{12 \times 3}{60} = \frac{36}{60}$

$= 0,6$

assumption:

we have 6 cases of counting: $F_3 F_2 F_1$

⋮

Exercise 08:

1. $p_n = \underbrace{\left(\frac{364}{365} \times \frac{364}{365} \times \cdots \times \frac{364}{365}\right)}_{n \text{ person}} = \left(\frac{364}{365}\right)^n$

2. $\overline{p_n} = 1 - p_n$

$$\overline{p_n} \geq 0,5 \implies 1 - p_n \geq 0,5$$

$$1 - \left(\frac{364}{365}\right)^n \geq 0,5 \implies \left(\frac{364}{364}\right)^n \leq 0,5$$

$(\ln x)$ is increasing:

$$n \ln\left(\frac{364}{365}\right) \leq \ln 0,5$$

$$n \geq \frac{\ln 0,5}{\ln\left(\frac{364}{365}\right)} \implies n \geq 252,64$$

$n = 253.$

ex 9

1) $|\Omega| = \binom{20}{6}\binom{14}{4}\binom{10}{5}\binom{5}{5} = 9777287520$

$= \frac{20!}{6!\,4!\,5!\,5!}$

2) $P(E) = \frac{\binom{4}{4}\binom{16}{2}\binom{14}{4}\binom{10}{5}\binom{5}{5}}{\binom{20}{6}\binom{14}{4}\binom{10}{5}\binom{5}{5}} = \frac{22}{1615} = 0,17$

ex 10

① E_1 XYZ E_3: YXZ E_5: ZXY

E_2 XZY E_4 YZX E_6: ZYX

$A: \{E_1, E_2, E_5\}$; $B\{E_1, E_2\}$; $C\{E_3, E_5\}$; $D\{E_4, E_6\}$

② A and B independent $\Leftrightarrow P(A)\,P(B) = P(A\cap B)$

* $P(A) = \frac{3}{6} = \frac{1}{2}$

$P(B) = \frac{2}{6} = \frac{1}{3}$ $P(A)\,P(B) = \frac{1}{6} \neq P(A\cap B)$

$P(A\cap B) = \frac{2}{6} = \frac{1}{3}$ they are not independent.

* $P(C) = \frac{2}{6} = \frac{1}{3}$ $P(A)\,P(C) = \frac{1}{6} = P(A\cap C)$

$P(A\cap C) = \frac{1}{6}$ they are independent.

* $P(D) = \frac{2}{6} = \frac{1}{3}$ $P(A)\,P(D) = \frac{1}{6} \neq P(A\cap D)$

$P(A\cap D) = 0$ they are not independent (disjoint).

$P(A/B) = P(A) \Rightarrow$ A and B are independent.

ex 11

- $P(A/B) = \frac{P(A\cap B)}{P(B)} = \frac{0,1}{0,3} = \frac{1}{3} = 0,33$
- $P(B/A) = \frac{P(B\cap A)}{P(A)} = \frac{0,1}{0,5} = \frac{1}{5} = 0,2$

$A \subset (A\cup B)$

$\hookrightarrow P(A\cap(A\cup B)) = P(A)$

- $P(A/A\cup B) = \frac{P(A\cap(A\cup B))}{P(A\cup B)} = \frac{P((A\cap A)\cup A\cap B)}{P(A\cup B)} = \frac{P(A\cup(A\cap B))}{P(A\cup B)}$

$\hookrightarrow P(A\cup(A\cap B)) = P(A) + P(A\cap B) - P(A\cap(A\cap B))$

$P(A\cup B) = P(A) + P(B) - P(A\cap B)$ $= P(A) + \cancel{P(A\cap B)} - \cancel{P(A\cap B)} = P(A)$

$= 0,5 + 0,3 - 0,1 = 0,7$

$\Rightarrow P(A/A\cup B) = \frac{P(A)}{P(A\cup B)} = \frac{0,5}{0,7} = 0,7 = \frac{5}{7}$

- $P(A/A\cap B) = \frac{P(A\cap A\cap B)}{P(A\cap B)} = \frac{P(A\cap B)}{P(A\cap B)} = 1.$

- $P(A\cap B/A\cup B) = \frac{P((A\cap B)\cap(A\cup B))}{P(A\cup B)} = \frac{P((A\cap B\cap A)\cup(A\cap B\cap B))}{P(A\cup B)} = \frac{P((A\cap B)\cup(A\cap B))}{P(A\cup B)} = \frac{P(A\cap B)}{P(A\cup B)}$

$(A\cap B) \subset (A\cup B)$

$\Rightarrow P((A\cap B)\cap(A\cup B)) = P(A\cap B)$

$= \frac{0,1}{0,7} = \frac{1}{7} = 0,14$

Exercise 12

1. $P(A).P(B^c) = P(A).(1-P(B))$
$= P(A) - P(A).P(B)$
$= P(A) - P(A \cap B)$
$= P(A \setminus B)$
$= P(A \cap B^c)$

$\Rightarrow A \perp\!\!\!\perp B$

$\Updownarrow$

$P(A^c \cap B^c) = P(A \cup B)^c$
$= 1 - P(A \cup B)$
$= 1 - [P(A) + P(B) - (P(A \cap B)]$
$= 1 - P(A) - P(B) + P(A \cap B)$
$= P(A^c) \times P(B^c)$.

mutually exclusive = disjoint

$A \perp\!\!\!\perp B \Leftrightarrow P(A \cap B) = P(A).P(B)$

2.
A and P disjoint
$\Rightarrow P(A \cap B) = 0$
$\Rightarrow P(A \cup B) = P(A) + P(B)$
$= 0{,}7 + 0{,}4$
$= 1{,}1$ impossible
because $0 \leq P \leq 1$

$P(A) + P(B)$ disjoint
$\Rightarrow P(A \cup B) = P(A) + P(B) =$
$= 0{,}4 + 0{,}3$
$= 0{,}7$

Yes possible.

Exercise 13:

Method 1:

A: the patient respond

$P(A) = 0{,}9$

$P(\bar{A}) = 0{,}1$

B: None respond

$P(B) = P(A) \times P(A) \times P(A)$.
$= (0{,}9)^3$
$= 0{,}999$.

(Because the three patients are independent).

Method 2:

$P(1) = \binom{3}{3} \times (0{,}9)^3 \times (0{,}1)^0 =$

$P(2) = \binom{3}{2} \times (0{,}9)^2 \times (0{,}1)^1 =$

$P(3) = \binom{3}{1} \times 0{,}9 \times (0{,}1)^2 =$

$P(\ \ ) = P(1) + P(2) + P(3)$
$= 0{,}999$.

Law of total probability:

$P(B) = P(B|A)\,P(A)$
$\quad + P(B|\bar{A})\,P(\bar{A})$.

S: Selected potential customer will purchase the product.

A: Seeing Ads.

$\bar{A}$: Not seeing Ads

<u>we have:</u>

M: Seeing magazine ads.

T: " TV ads.

$P(M) = \frac{1}{50}$

$P(T) = \frac{1}{5}$

$P(M \cap T) = \frac{1}{100}$

$P(S|_A) = \frac{1}{3}$

$P(S|_{\bar{A}}) = \frac{1}{10}$

$P(A) = P(M) + P(T) - P(M \cap T)$.

$P(A) = \frac{1}{50} + \frac{1}{5} + \frac{1}{100}$

$P(A) = \frac{2}{100} + \frac{20}{100} - \frac{1}{100} = \frac{21}{100}$

$P(\bar{A}) = 1 - P(A) = 1 - \frac{21}{100} = \frac{79}{100}$

A, $\bar{A}$ are partition of Ω, so we can apply the law of total probability:

$P(S) = \frac{21}{100} \times \frac{1}{3} + \frac{79}{100} \times \frac{1}{10}$

$= \frac{149}{100}$

EXO 17

A : student know, B : student guess

$P(B|\bar{A}) = 0{,}25, \quad P(B|A) = 1.$

$$P(A|B) = \frac{P(B|A).P(A)}{P(B|A).P(A) + P(B|\bar{A}).P(\bar{A})}$$

$$= 0.94$$

EXO 18

A : two ball white

B_i : selected from i

A_i : selected from i

$P(A_i|B_i) = \binom{i}{2} / \binom{5}{2} =$ [illegible]

$$P(A) = \sum_{i=1}^{5} P(A_i|B_i)\,P(B_i) = 0{,}4$$

A : The Two balls are white.

B_i : is selected bowls i, $P(B_i) = 1/5$

A_i : the two selected balls are from bowl i

$$P(A_1 | B_1) = \frac{\binom{1}{2}}{\binom{5}{2}} = \frac{\binom{1}{2}}{10}$$

$$P(A) = \sum_{i=1}^{5} P(A_i | B_i) \cdot P(B_i) = \left(\frac{1}{5}\right) 0 + \frac{1}{5}\frac{1}{10} + \frac{1}{5}\frac{3}{10} + \frac{1}{5}\frac{6}{10} + \frac{1}{5}$$

$$= 0,4$$

$$P(A/B_3) = \frac{P(A \cap B_3)}{P(B_3)}$$

$$P(B_3 \cap A) = \underline{P(A/B_3)}\, P(B_3)$$

$$\times \frac{1}{5}$$

ex 19 :

we have a fair coin $\Rightarrow p(H) = p(T) = 1/2$

The experiment stops when head appears or 5 tosses are made

$X = x_i$	1	2	3	4	5
$p(X = x_i)$	$1/2$	$1/$	$1/8$	$1/16$	$1/16$

$\Sigma = 1$

$p(X = 1) = p(h) = 1/2$

$p(X = 2) = p(\{Th\}) = (1/2)^2 = 1/4$

$p(X = 3) = p(\{TTh\}) = (1/2)^3 = 1/8$

$p(X = 4) = p(\{TTTh\}) = (1/2)^4 = 1/16$

$p(X=5) = p(\{TTTTT\}) = 1 - \sum_{i=1}^{4} p(X=x_i) = 1/16$

ex 20 :

1- The CDF of x:

$$\text{CDF} : F(x) = \begin{cases} 0 & / \; x < 77 \\ 0{,}15 & / \; 77 \leq x < 78 \\ 0{,}3 & / \; 78 \leq x < 79 \\ 0{,}5 & / \; 79 \leq x < 80 \\ 0{,}9 & / \; 80 \leq x < 81 \\ 1 & / \; 81 \leq x \end{cases}$$

2- $p(X=80) = 0{,}4$.

$p(X > 80) = p(X=81) = 0{,}1$

$p(X \leq 80) = p(X=81) = 0{,}9$

3- $\mu_x = \sum x_i \, p(X=x_i)$

$= 77 \times 0{,}15 + 78 \times 0{,}15 + 79 \times 0{,}2 + 80 \times 0{,}4 + 81 \times 0{,}1$.

$\mu_x = 79{,}15$.

• $\sigma_x^2 = E(X^2) - E(X)^2$

$= \sum x_i^2 \, p(X=x_i) - \mu_x^2$

$= 1{,}5275$

ex 21 :

1- The probability distribution of X:

$p(X=0) = 6/36$

$p(X=1) = 10/36$

$p(X=2) = 8/36$

$p(X=3) = 6/36$

$p(X=4) = 4/36$

$p(X=5) = 2/36$

- CDF of x:

$F(0) = \frac{6}{36}$, $F(3) = \frac{30}{36}$

$F(1) = \frac{16}{36}$, $F(4) = \frac{34}{36}$

$F(2) = \frac{24}{36}$, $F(5) = 1$.

3- The mean:

$\mu_x = E(x) = \sum x_i \, p(X=x_i)$

$= 0 \cdot \frac{6}{36} + 1 \cdot \frac{10}{36} + 2 \cdot \frac{8}{36} + 3 \cdot \frac{6}{36} + 4 \cdot \frac{4}{36}$

$+ 5 \cdot \frac{2}{36} = 70/36$

$\sigma_x^2 = E(X^2) - EX^2$

$= \sum x_i^2 \, p(X=x_i) - \left(\frac{70}{36}\right)^2$

$= \frac{10}{36} + 4 \cdot \frac{8}{36} + 9 \cdot \frac{6}{36} + 16 \cdot \frac{4}{36} + 25 \cdot \frac{2}{36} - ($